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A Tutorial on Quantum Key Distribution Protocols With Interactive Simulation of QBER and SKR Performance Metrics

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ABSTRACT

In recent years, Quantum Key Distribution (QKD) has experienced rapid advancements, leading to widespread practical applications and drawing considerable attention from both academic and commercial sectors. Consequently, there has also been a growing interest in investigating the performance and efficiency of various QKD protocols. In this tutorial paper, we present a theoretical simulation-based review of several QKD protocols, such as BB84, Decoy State BB84, Coherent One Way (COW), Differential Phase Shift (DPS), and the entanglement-based BBM92 and E91 protocols. We simulate the Quantum Bit Error Rate (QBER) and Secret Key Rate (SKR) under varying parameters to provide a deeper understanding of their performance and inner workings. In addition to the simulation results, we introduce a web-based graphical user interface (GUI) that allows users to interactively explore protocol behaviour and visualise the impact of various system parameters on the QBER and SKR of the QKD protocols. This tutorial aims to offer an intuitive understanding of various QKD protocols from an engineering perspective, using accessible simulation and visual analysis tools.

1 | Introduction

The use of communication systems has become integral in our modern-day-to-day life. The security of such systems typically depends on classical asymmetric cryptographic algorithms like the RSA. In such algorithms, we assume a bounded adversary, that is, we assume attackers have limited computational power, making certain mathematical problems (like the factorisation of large numbers or the discrete logarithm problem) infeasible to solve. Unfortunately, if the bounds fail, either through algorithmic breakthrough or the use of quantum computers, our current cryptographic infrastructure will collapse [1]. Thus, there is an urgent need for a solution to safeguard against quantum attacks, and one of the most prominent approaches is quantum cryptography. Quantum cryptography ensures security against

computationally unbounded adversaries by exploiting the laws of quantum physics [2], such as the quantum no-cloning theorem [3] and the Heisenberg uncertainty principle [4].

One of the most widely studied and successfully applied techniques in quantum cryptography is the Quantum Key Distribution (QKD) protocols [5], which aim to securely generate a shared key between two parties (Alice and Bob) over a public channel, without information leaking to an eavesdropper (Eve). This is achieved through utilising quantum mechanical properties to encode and share the secret key.

Beyond academic research, efforts are also being directed towards realising QKD systems suitable for deployment in real communication networks. Various fibre-based QKD networks

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have already been deployed in the field [2], such as the DARPA Quantum Network [6], the Cambridge Quantum Network [7], and the Swiss Quantum Network [8], etc.

Funding for quantum technologies has continued to rise in recent years. In 2024, investors contributed nearly \$1.59 billion to quantum technology start-ups worldwide, specifically quantum software investments also showed remarkable growth in 2024, reaching \$621 million [9]. Similarly, public funding by several governments around the world has also increased (US, France, Europe, India), around \$44.5Bn has also been committed to research and infrastructure development through several national grants and budgets [9–11]. As QKD systems undergo widespread adoption, there is an increasing need to comprehend the underlying phenomena and understand their comparative efficiency under different conditions. While several prior works have explored quantum simulators and QKD protocols, our emphasis here is on education and accessibility through this tutorial and its accompanying toolkit.

In this tutorial, our focus is on BB84-based protocols, including standard BB84, Decoy State, Coherent One Way (COW), and Differential Phase Shift (DPS) QKD schemes, alongside entanglement-based protocols such as E91 and BBM92. By systematically examining these protocols under asymptotic conditions, we focus on analysing and validating fundamental theoretical relationships between key performance metrics such as quantum bit error rate (QBER) and secret key rate (SKR). Furthermore, we introduce an interactive web-based graphical user interface (GUI) designed to improve accessibility and learning. The platform allows users to experiment with parameters, such as channel attenuation, and immediately observe how these changes affect critical quantities like QBER and SKR.

Different QKD protocols have been developed for different practical communication needs. The BB84 protocol is the most widely implemented QKD scheme and has been deployed in real optical fibre networks. Large metropolitan QKD networks have demonstrated stable BB84 operation over installed fibre infrastructure [12]. In practical systems, weak coherent pulses are used instead of ideal single-photon sources. To maintain security in such realistic implementations, the decoy-state BB84 protocol is applied. Decoy-state QKD has enabled long-distance fibre transmission [13] and is now standard in commercial QKD deployments. The Coherent One-Way (COW) protocol is designed for high-speed fibre-based communication with simple hardware requirements and has been experimentally demonstrated in deployed fibre links [14]. The Differential Phase Shift (DPS) protocol encodes key bits in the phase difference between consecutive pulses and has been demonstrated over long optical fibre distances as well [15]. Entanglement-based protocols such as E91 and BBM92 are important for quantum networks and advanced security models. Entanglement-based QKD has been experimentally demonstrated over satellite links over long distances [16], showing its suitability for long-distance global quantum communication.

To further enhance the readers' understanding of QKD protocols and explore these metrics, this tutorial mainly aims to offer the following insights and tools:

- i. Present a comprehensive review of QKD protocols, including a detailed survey of their key performance metrics like the QBER and SKR.
- ii. Introduce a theoretical simulator toolkit to assess the performance of QKD protocols under the fibre-based and free-space optical channels.
- iii. Familiarise the readers with the toolkit's modelling and practical factors that affect real-world QKD implementations, such as channel loss, detector inefficiency, dark counts, various types of noises, etc.
- iv. Present a web-based graphical interface where users can interact, adjust system parameters and visualise how these changes affect QKD performance.
- v. Serve as a practical, hands-on learning resource for students and researchers, particularly those new to quantum cryptography, by introducing QKD from an engineering perspective.

The rest of the paper is organised as follows. In Section 2, we begin by reviewing some of the related works to establish the background and the motivation of this paper. In Section 3, we discuss the fundamental physical components of a QKD system, namely the source, channel, and detector, which form the building blocks for simulating all the subsequent protocols. In Section 4, we present the definitions and significance of QBER and SKR, setting the foundation for analysing their behaviour in different QKD protocols. Next, in Section 5, we provide an introduction to some of the most common prepare-and-measure (BB84, Decoy State, COW, DPS) and entanglement-based (BBM92, E91) QKD protocols, with particular emphasis on their QBER and SKR performance metrics. In Section 6, numerical simulations are presented through QBER and SKR graphs, illustrating the impact of key parameters such as distance and mean photon number on the performance of various QKD protocols. Furthermore, in Section 7, we introduce a web-based GUI that allows users to visualise and interact with the results of these protocols. Finally, in Section 8, we conclude with a summary of our findings and potential directions for future work.

2 | Related Works

QKD has grown significantly in the last few decades, and there has also been a lot of interest in simulating these protocols for cost-effective research and development. The main challenge lies in the fact that each QKD protocol is distinct and typically requires specialised hardware, hence they can be implemented by experts [20]. Numerous researchers have attempted to create simulation frameworks that alleviate this problem through various approaches. This section reviews the various simulation approaches and gives a general overview of the techniques used. Furthermore, Table 1 gives a concise chronological summary of the existing simulation frameworks.

The most common approach is to directly simulate mathematical equations describing the protocol [17, 21]. Other works in this area also build on this, with additions like incorporating classical, event-based simulation approaches [19] or abstracting

TABLE 1 | Categorisation of QKD simulation works.

Category	Author(s)	Paper name	Year
Mathematical simulation	Ma et al. [17]	Practical decoy state for quantum key distribution	2005
	Lo et al. [18]	Decoy state quantum key distribution	2005
	Zhao & Raedt [19]	Event-by-event simulation of quantum cryptography protocols	2007
	Coles et al. [20]	Numerical approach for unstructured quantum key distribution	2016
	Zhang et al. [21]	Improved key rate bounds for practical decoy-state quantum key distribution systems	2017
	Primaatmaja et al. [22]	Discrete-variable quantum key distribution with homodyne detection	2022
	Mizutani et al. [23]	Differential-phase-shift QKD with practical Mach–Zehnder interferometer	2024
	Sandfuchs et al. [24]	Security of differential phase shift QKD from relativistic principles	2025
	Hui & Chen [25]	Simulation of BB84 quantum key distribution in depolarising channel	2009
	Jasim et al. [26]	Quantum key distribution: Simulation and Characterisations	2015
Software-based physical simulation	Khalid [27]	QuCCs: An experimental of quantum key distribution using quantum cryptography and communication simulator	2019
	Lardier et al. [28]	Quantum-Sim: An open-source Co-simulation platform for quantum key distribution-based smart grid communications	2019
	Chatterjee et al. [29]	qkdSim, a simulation toolkit for quantum key distribution including imperfections: Performance analysis and demonstration of the B92 protocol using heralded photons	2020
	Khan et al. [30]	Security analysis of QKD protocols: Simulation & comparison	2020
	Abdelgawad et al. [31]	EnQuad: A publicly-available simulator for quantum key distribution protocols	2020
	Caputo et al. [32]	A simulator of optical coherent-state evolution in quantum key distribution systems	2022
	Sethia & Banerjee [33]	A MATLAB based modelling and simulation package for DPS-QKD	2022
	Adu-Kyere et al. [34]	Quantum key distribution: Modelling and simulation through BB84 protocol using Python3	2022
	Gkoularas et al. [35]	NuQKD: A modular quantum key distribution simulation framework for engineering applications	2024
	Pereszlenyi [36]	Simulation of quantum key distribution with noisy channels	2005
Platform-based simulator	Buhari et al. [37]	An efficient modelling and simulation of quantum key distribution protocols using OptiSystem	2012
	Sorensen & Grimaila [38]	Discrete event simulation of the quantum channel within a quantum key distribution system	2015
	Mailloux et al. [39]	A modelling framework for studying quantum key distribution system implementation Nonidealities	2015
	Fan-Yuan et al. [40]	A universal simulating framework for quantum key distribution systems	2020
	Mallick et al. [41]	Long distance QKD propagation using optical single sideband scheme	2024
	Abhignan et al. [42]	Simulations of distributed-phase-reference quantum key distribution protocols	2024
	Mallick et al. [43]	Secure real-time transmission of multi-spectral satellite images inducing a 6D hyper-chaotic system and BB84 QKD protocol	2025

(Continues)

TABLE 1 | (Continued)

Category	Author(s)	Paper name	Year
Education & visualisation simulators	Niemiec et al. [44]	Quantum cryptography protocol simulator	2011
	Kohnle & Rizzoli [45]	Interactive simulations for quantum key distribution	2017
	Nurhadi & Syambas [46]	Quantum key distribution (QKD) protocols: A survey	2018
	Amer et al. [1]	An introduction to practical quantum key distribution	2021

the protocol and parameters involved [20]. Apart from theoretical analysis and mathematical simulations, there have also been several software-based simulators that focus on replicating the physical components, communication architectures, and protocol-specific parameters using code, making them far more suitable for real-world testing and evaluations. Various MATLAB-based simulators have been developed for modelling QKD systems [25, 29, 31–33], while others have adopted multi-stack architectures [26, 27, 34, 35], integrating multiple simulation layers (e.g., physical, network, and application) to provide a more comprehensive and realistic representation of QKD networks. As a subset of the previous category, there are works that build upon already existing photonics software platforms like OMNet++ or ANSYS, for a more comprehensive frameworks that support the simulation of multiple QKD protocols [28, 30, 36–39, 42].

A lot of work has been done in the field of simulating QKD protocols to ease the burden associated with specialised hardware requirement and optimisation. Despite numerous studies on QKD schemes, there is a lack of a comprehensive survey of various QKD protocols in terms of their QBER and SKR metrics. This tutorial aims to bridge the gap between physics-oriented QKD protocols and beginners by presenting a survey of QKD protocols and their performance metrics. This tutorial will enable newcomers to have a thorough understanding of QKD protocols and conduct future research on the integration of QKD with future communication networks. Furthermore, only a few studies have addressed the important area of educational and visualisation tools aimed at improving the understanding of QKD protocols for newcomers to the field. Some works exist [1, 44–46], which try to give a holistic and intuitive understanding of QKD through hands-on interaction, but there is a significant gap between the advancements and the corresponding educational and simulation tools available. In continuation of this type of work, our tutorial further aims to be a soft launching pad for those who are new to QKD protocols from an engineering perspective.

Unlike existing simulators and toolkits, our educational, web-accessible framework allows easy comparison of multiple QKD protocols under consistent assumptions. Table 2 positions our contribution against prior simulator categories (columns) across key capabilities (rows), indicating support levels (✓: full, Δ: partial, ×: none). The column categories (already mentioned in Table 1) are defined as: **Mathematical Simulations**, which strictly derive theoretical key rates and security proofs; **Software-Based Physical Simulations**, which use programming (e.g., Python, MATLAB) to model hardware imperfections and discrete quantum events; **Platform-Based** frameworks, which integrate QKD into existing software platforms (e.g., OptiSystem,

OMNet++); and **Educational/Visualisation Simulators**, which prioritise pedagogical interaction over simulation. As shown, **This Work** bridges these domains by combining rigorous multi-protocol analysis with an interactive educational GUI.

3 | Physical Components

This section describes the simulation of various physical components used in the QKD protocols, such as the source, channel and detector. These components form the backbone of any QKD system, as they determine how quantum states are generated, transmitted, and measured. Accurately modelling these elements allows us to capture real-world imperfections and understand their impact on system performance and security. The models developed here serve as the foundation for the different QKD protocols discussed throughout this paper.

3.1 | Source

For prepare and measure-based QKD protocols, such as BB84, Decoy State, COW, and DPS QKD, we model the single photon sources as Weak Coherent States (WCS) which are easier to implement in practice. These protocols follow a Poisson photon number distribution. For a WCS with mean photon number μ , the probability of generating exactly n photons in a pulse is given by the Poisson distribution:

$$P(n) = \frac{\mu^n e^{-\mu}}{n!} \quad (1)$$

On the other hand, entanglement-based QKD protocols such as BBM92 and E91 model the photon source as a type-II Parametric Down-Conversion (PDC) source. The quantum state emitted by such a source is described by a two-mode squeezed vacuum state:

$$|\Psi\rangle = (\cosh \chi)^{-2} \sum_{n=0}^{\infty} \sqrt{n+1} (\tanh \chi)^n |\Phi_n\rangle \quad (2)$$

where the n -photon pair component is given by

$$|\Phi_n\rangle = \frac{1}{\sqrt{n+1}} \sum_{m=0}^n (-1)^m |n-m, m\rangle_a |m, n-m\rangle_b \quad (3)$$

For $n = 1$, this expression reduces to the maximally entangled Bell state used in E91 and BBM92 protocols. From the expansion coefficients of the two-mode squeezed state, the probability

TABLE 2 | Capability comparison of existing QKD simulators and the proposed framework.

Feature	Mathematical simulation [17–24]	Software-based physical simulation [25–35]	Platform based [36–43]	Edu./vis. simulators [1, 44–46]	This work
Multi-protocol comparison	Δ	✓	✓	Δ	✓
Entanglement + P&M protocols	×	Δ	Δ	×	✓
Unified QBER/SKR analysis	✓	Δ	Δ	×	✓
Web-based GUI accessibility	×	×	×	Δ	✓
Educational/tutorial focus	×	×	×	✓	✓
Interactive parameter exploration	×	Δ	Δ	✓	✓

of generating an n -photon pair is obtained by taking the squared modulus of the amplitude, yielding [47]:

$$P(n) = \frac{(n+1) \binom{\mu}{2}^n}{\left(1 + \frac{\mu}{2}\right)^{n+2}} \quad (4)$$

where $\lambda = \sinh^2 \chi$ and the expected photon-pair number is $\mu = 2\lambda$, which represents the mean photon-pair number per pulse and characterises the brightness of the PDC source. These photon statistics are used in the simulator to compute photon-number dependent gains, losses, and multi-photon contributions, which are essential for accurately modelling the security behaviour of entanglement-based QKD protocols.

3.1.1 | Quantum States of the System

To further understand the impact of sources, it is necessary to define the quantum states describing photons:

3.1.1.1 | Polarisation States. Polarisation encodes qubits, in protocols such as BB84, using the polarisation degree of freedom of a single photon. The Z basis (computational) is given by horizontal and vertical polarisations:

$$|0\rangle \equiv |H\rangle, \quad |1\rangle \equiv |V\rangle \quad (5)$$

The X basis (diagonal) consists of the superpositions of the Z-basis states:

$$|+\rangle = \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle) \quad |-\rangle = \frac{1}{\sqrt{2}}(|H\rangle - |V\rangle) \quad (6)$$

The Y basis (circular) also uses superpositions, but with a complex phase:

$$|R\rangle = \frac{1}{\sqrt{2}}(|H\rangle + i|V\rangle) \quad |L\rangle = \frac{1}{\sqrt{2}}(|H\rangle - i|V\rangle) \quad (7)$$

The prepare and measure protocols typically use two conjugate bases, such as the Z-basis and the X-basis. When Alice randomly prepares qubits in one of these bases and Bob randomly measures in one of them, the laws of quantum mechanics ensure that any eavesdropping attempt will introduce detectable errors.

3.1.1.2 | Bell States. Entanglement-based QKD protocols rely on maximally entangled two-photon states, known as Bell states:

$$\begin{aligned} |\Phi^+\rangle &= \frac{1}{\sqrt{2}}(|HH\rangle + |VV\rangle), & |\Phi^-\rangle &= \frac{1}{\sqrt{2}}(|HH\rangle - |VV\rangle) \\ |\Psi^+\rangle &= \frac{1}{\sqrt{2}}(|HV\rangle + |VH\rangle), & |\Psi^-\rangle &= \frac{1}{\sqrt{2}}(|HV\rangle - |VH\rangle) \end{aligned} \quad (8)$$

Here, the first ket corresponds to Alice's photon and the second to Bob's. These states exhibit nonlocal correlations, and the violation of Bell inequalities forms the basis for the security guarantees of entanglement-based QKD. The E91 protocol, for example, leverages the strong correlations of a Bell state, such as the anti-correlated singlet state $|\Psi^-\rangle$, to detect eavesdropping.

3.2 | Channel

In our simulation framework, we model the quantum channel in two modes, optical fibre-based and free space optical-based. The total transmission efficiency (η_{ch}) depends on the transmission mode and the transmission distance (d) between Alice and Bob.

3.2.1 | Fibre-Based Model

In the optical fibre-based channel model, the loss is modelled using exponential attenuation based on the attenuation coefficient (α) in decibels per kilometre (dB/km). Here, we also consider a base efficiency (η_0) of the channel, in practical scenarios, it can be adjusted to account for other sources of inefficiency such as imperfect coupling between optical components, connector losses, or misalignment effects. Thus, the effective transmission efficiency is:

$$\eta_{\text{ch}}^{\text{fibre}}(d) = \eta_0 10^{-\alpha d/10} \quad (9)$$

3.2.2 | Free Space Optical (FSO) Model

For the FSO channel mode, the total transmittance efficiency is a product of various attenuation factors due to physical and environmental losses in the FSO path. These include geometrical

spread (η_{geo}) and atmospheric loss (η_{atm}). Thus the effective transmission efficiency is [48]:

$$\eta_{\text{ch}}^{\text{fso}}(d) = \eta_0 \eta_{\text{geo}} \eta_{\text{atm}} \quad (10)$$

The geometrical loss is the decrease in optical power as the beam travels from the transmitter to the receiver, primarily due to the beam's natural divergence. It can be determined by the divergence of the beam (θ) and the ratio of the receiving aperture size to the beam footprint at the receiver. It is given by:

$$\eta_{\text{geo}} = \frac{d_r^2}{(d_t + d\theta)^2} \quad (11)$$

where d_r, d_t represent the diameter of the receiver and transmitter, respectively. One of the major sources of attenuation in FSO communication is the atmospheric loss, which arises due to absorption and scattering by air molecules, aerosols, and other particles along the propagation path [49]. We have modelled this using the Beer-Lambert law, which describes the exponential attenuation of an optical signal as it travels through a medium with absorbing or scattering properties. Thus, the atmospheric loss is [50]:

$$\eta_{\text{atm}} = e^{-\alpha_{\text{atm}} d} \quad (12)$$

where, α_{atm} is the atmospheric attenuation coefficient.

3.3 | Detector

We model the single-photon detectors (SPDs) using fundamental physical parameters. The detector efficiency (η_{det}) is defined as the probability that an incident photon is successfully registered by the detector [5]. The total end-to-end transmittance (η) from Alice to Bob is often given by:

$$\eta = \eta_{\text{ch}} \eta_{\text{det}} \quad (13)$$

Secondly, dark count rate (δ) is defined as the average rate of spurious detection events (noise) in the absence of photons, typically modelled as a Poisson process [51]. The probability of at least one dark count occurring within a finite detection window (τ) is:

$$P_{\text{dark}} = 1 - e^{-\delta\tau}. \quad (14)$$

For a given number of incoming photons n , the probability of detection is given by:

$$P_{\text{detect}}(n) = 1 - (1 - \eta_{\text{det}})^n \quad (15)$$

The models described above serve as a base version used consistently across all protocols in our simulation framework. However, each protocol may incorporate additional refinements or protocol-specific modifications to better capture its unique characteristics or error sources as needed. Additionally, in the numerical simulation, detector responses are treated as stochastic processes. Detection events are not assigned

deterministically; instead, the calculated detection probabilities are used to randomly sample whether a click occurs for each pulse. Similarly, dark counts are generated probabilistically according to a Poisson model within the detection time window. The parameters listed in Table 4 therefore define the underlying probability distributions governing detector behaviour rather than fixed detection outcomes. This approach allows the simulator to capture the intrinsic randomness of single-photon detection processes.

4 | Key Performance Matrix

To evaluate the performance and security of any QKD system, it is essential to define measurable parameters that reflect how well the system operates under realistic conditions. Among these, the most important are the QBER, which measures the accuracy of key transmission, and the SKR, which determines how efficiently secure keys can be generated. Together, these parameters provide a comprehensive understanding of the trade-offs between performance, noise tolerance, and security across different QKD protocols.

4.1 | QBER

QBER is the measure of how noisy or unreliable the communication channel is in a QKD system. It essentially tells us what portion of the received bits were incorrect (ϵ_{total}) compared to what was originally sent (G_{total}). It can be mathematically formulated as :

$$Q = \frac{\epsilon_{\text{total}}}{G_{\text{total}}} \quad (16)$$

The QBER gets even more useful because it can also be used as an indicator of the presence of an eavesdropper Eve, because by attempting to extract knowledge, Eve inherently adds noise. There is a direct relation between the amount of information extracted by the Eve and the amount of noise in the system. Other than the errors introduced by the eavesdropper, a major source is the imperfections in the system itself. To identify and separate these effects, it is even more important to study the effects of QBER as a key metric. QBER can reflect errors introduced by random noise (such as detector dark counts, electronic noise, or afterpulsing), optical imperfections (like misalignment, polarisation or phase errors, and multi-photon pulses), channel disturbances (losses in fibres or free space, timing jitter, and environmental variations), and device limitations (including detector inefficiency and crosstalk).

4.2 | SKR

The SKR is the rate at which Alice and Bob can generate secure bits after accounting for errors and potential eavesdropping. The SKR essentially tells us how many bits per signal can be distilled into a secure key. A general expression of SKR is given by the Devetak-Winter bound [52]:

$$SKR = I_{AB} - \chi_{AE} \quad (17)$$

where I_{AB} is the classical Shannon mutual information between Alice and Bob, and χ_{AE} is the quantum Holevo information, representing the maximum information Eve can obtain about Alice's key. Intuitively, this means the final secure rate is the portion of information Alice and Bob share minus the information that could have leaked to Eve. In practice, this bound is realised through two key processes:

- i. Error Correction: Bits that were detected incorrectly due to an error and must be corrected, reducing the final key length.
- ii. Privacy Amplification: Information that may have been leaked to Eve, thus additional bits must be removed to ensure secrecy.

The SKR depends on the QBER, a higher QBER means more errors to correct and a greater risk of information leakage, which lowers the amount of secure key that can be distilled. In the following, we explore these key parameters in different QKD protocols, showing how they influence the overall performance and security.

5 | Overview of Protocols

This section provides an overview of the various QKD protocols. Each protocol follows the same fundamental principle of using quantum mechanics to establish a shared secret key between two parties, but they differ in how quantum states are prepared, transmitted, and measured. We cover both prepare-and-measure schemes, such as BB84, Decoy State, COW, and DPS, and entanglement-based schemes, such as E91 and BBM92. By studying these protocols under common constraints, we can compare their performance, understand their advantages and weaknesses, and see how different design choices influence the key rate and security.

5.1 | BB84 QKD Protocol

BB84 QKD protocol, developed by Bennett and Brassard [53], is one of the most studied discrete variable quantum key distribution (DV-QKD) protocols. The BB84 protocol mainly consists of the transmitter Alice, the receiver Bob, the quantum channel, and the authenticated classical channel. The protocol proceeds as follows:

- i. Alice generates and encodes random bits or photon pulses using random bases or the polarisation orientation of the electromagnetic wave.
- ii. Two mutually non-orthogonal bases are used: the rectilinear (Z) basis or the diagonal (X) basis.
 - If the Z basis is selected, key bit 0 is encoded as a horizontally polarised pulse with state vector $|H\rangle$, and key bit 1 is encoded as a vertically polarised pulse $|V\rangle$.
 - If the X basis is selected, key bit 0 is encoded as $|+\rangle$, and key bit 1 is encoded as $|-\rangle$.

- iii. These encoded bits are then transmitted to Bob through the quantum channel.
- iv. Bob measures the incoming photons using a randomly chosen basis.
- v. Alice and Bob then announce and compare their measurement bases over an authenticated classical channel, that is Eve can listen but cannot alter this discussion.
- vi. They keep only the measurement outcomes where their bases matched. These form the *sifted key*.
- vii. To check for possible eavesdropping and to estimate how much Eve might have learnt, they use a subset of the sifted bits to calculate the error rate (QBER).
- viii. If the error rate is within acceptable limits, Alice and Bob apply classical error correction to fix mistakes.
- ix. They then perform privacy amplification to remove any residual information that Eve may still have.
- x. As a result, they distil a smaller but highly secure key that is indistinguishable from a truly random sequence, protecting it against significant information leakage.

5.1.1 | BB84 QBER Analysis

In this section, we describe the process of calculating the QBER and other metrics related to the BB84 QKD protocol. In continuation of our detector framework, we model some more errors and effects for a more robust simulation. For multi-photon pulses ($n > 1$), the detector saturation effect (η_s) is modelled using a correction factor ($\beta \approx 0.02$) to account for crosstalk and other nonlinearities due to multiple photons, as follows [54]:

$$\eta_s = 1 + \beta(n - 1) \quad (18)$$

This saturation factor directly affects the detection probability and can therefore influence the QBER and, indirectly the SKR.

Additionally, our model incorporates afterpulsing effects with probability $\epsilon_{\text{after}} \approx 0.02$ [55] and timing jitter errors with probability $\epsilon_{\text{jitter}} \approx 0.01$ [56], which further contribute to the quantum bit error rate. The timing jitter represents the uncertainty in the detection timing, causing some photons to be registered outside their intended detection window. This causes a reduced fraction of correctly detected photons, thus affecting transmittance and QBER. Afterpulsing errors arise from trapped carriers, causing secondary detections after a real photon event due to trapped ion charge carriers in the detector. This increases the QBER because some detected bits are false, and consequently reduces the SKR. These error values are decided heuristically and can be further adjusted according to the user's needs. In order to calculate the QBER, we track two primary quantities, P_{correct} representing correct signal detections and ϵ representing erroneous signal detections [57], P_{correct} is given by:

$$P_{\text{correct}} = \sum_{n=1}^{n_{\text{max}}} P(n) \sum_{k=1}^n P_{\text{trans}}(n, k) P_{\text{detect}}(k) (1 - \epsilon_{\text{optical}}) \quad (19)$$

Similarly, the error probability becomes:

$$\epsilon = \sum_{n=1}^{n_{\max}} P(n) \sum_{k=1}^n P_{\text{trans}}(n, k) P_{\text{detect}}(k) \epsilon_{\text{optical}} \quad (20)$$

In the equations above, $P(n)$ represents the photon number distribution as defined in Equation (1). Also, here $\epsilon_{\text{optical}}$ captures the inherent optical errors in the system using the misalignment errors, jitter errors, and multi-photon errors:

$$\epsilon_{\text{optical}} = \epsilon_{\text{misalign}} + \epsilon_{\text{jitter}} + \epsilon_{\text{multi}} \quad (21)$$

Misalignment errors ($\epsilon_{\text{misalign}}$) are the probability that a photon will be incorrectly measured even when Alice and Bob use the same basis. This error arises from imperfect optical alignment between the quantum state preparation at the sender and the measurement at the receiver. Multi-photon error (ϵ_{multi}) occurs specifically when a pulse contains more than one photon. In our simulator, multi-photon error is modelled as increasing linearly with the number of extra photons in a pulse, using a small user-defined scaling factor ($\zeta \approx 0.02$) to capture the higher risk of errors in multi-photon states.

The transmission probability, denoted as $P_{\text{trans}}(n, k)$, represents the likelihood that k photons reach the receiver when n photons are sent through the quantum channel. This probability follows a binomial distribution based on the channel efficiency (η_{ch}):

$$P_{\text{trans}}(n, k) = \binom{n}{k} \eta_{\text{ch}}^k (1 - \eta_{\text{ch}})^{n-k} \quad (22)$$

Dark counts contribute to the quantum bit error rate because they generate random detection events unrelated to the transmitted signal. Since these random events do not correlate with the sent bits, they have a 50% probability of causing an error. However, dark count errors only contribute to the total error when no actual signal photon is detected. The dark count error is therefore calculated as:

$$\epsilon_{\text{dark}} = \frac{1}{2} P_{\text{dark}} (1 - P_{\text{correct}} - \epsilon) \quad (23)$$

Afterpulsing error (ϵ_{after}) occur when charge carriers get trapped in the detector material during the detection process and are released later, triggering secondary detections unrelated to incident photons. They are formulated as follows:

$$\epsilon_{\text{after}} = \frac{1}{2} (P_{\text{correct}} + \epsilon) P_{\text{after}} \quad (24)$$

Finally, we combine these errors into a single value, the total error (ϵ_{total}), which is given by:

$$\epsilon_{\text{total}} = \epsilon + \epsilon_{\text{dark}} + \epsilon_{\text{after}} \quad (25)$$

The QBER is finally calculated as the ratio of the total error probability to the total detection probability:

$$Q_{\text{BB84}} = \frac{\epsilon_{\text{total}}}{G} \quad (26)$$

The overall gain (G) represents the number of detection events (including both real detections and dark counts) per emitted pulse [58]. It can be calculated as:

$$G = P_{\text{signal}} + P_{\text{vac_dark}} + P_{\text{loss_dark}} \quad (27)$$

where,

$$\begin{aligned} P_{\text{detect}|n} &= \sum_{k=1}^n P_{\text{trans}}(n, k) P_{\text{detect}}(k) \\ P_{\text{signal}} &= \sum_{n=1}^{n_{\max}} P(n) P_{\text{detect}|n} \\ P_{\text{vac_dark}} &= P(0) P_{\text{dark}} \\ P_{\text{loss_dark}} &= \sum_{n=1}^{n_{\max}} P(n) (1 - P_{\text{detect}|n}) P_{\text{dark}} \end{aligned} \quad (28)$$

P_{signal} represents the probability of detecting at least one photon from all the incoming signals. $P_{\text{vac_dark}}$ signifies the dark counts when no photons were sent from the source, while $P_{\text{loss_dark}}$ represents the count when photons were sent but were lost, and a dark count nonetheless triggered a click. Together they make up G or the overall gain.

5.1.2 | BB84 SKR Analysis

Using the terms mentioned above, we can formulate the secret key rate expression for BB84 [59]:

$$\text{SKR} = \frac{1}{2} [G_1 (1 - H_2(\epsilon_1 Q_{\text{BB84}})) - G f_{\text{ec}} H_2(Q_{\text{BB84}})] \quad (29)$$

where G_1 is the gain for a single photon, that is the contribution to the raw key rate from pulses that contain exactly one photon and are successfully detected by the receiver. The gain is formulated as:

$$G_1 = P(1) P_{\text{detect}|n=1} \quad (30)$$

The binary entropy function $H_2(\cdot)$ represents the uncertainty or average information content of a binary random variable. It is defined as:

$$H_2(p) = -p \log_2(p) - (1 - p) \log_2(1 - p) \quad (31)$$

In Equation (29), the negative term corresponds to the information loss due to error correction. Error correction requires Alice and Bob to exchange some classical information to reconcile their keys. This reduces the amount of net secret key they can extract. The negative sign reflects this cost. Here, f_{ec} is the error correction factor, heuristically kept at $f_{\text{ec}} = 1.1$, in the context of the theoretical Shannon limit. The factor f_{ec} accounts for the practical reality that real error correction codes cannot

achieve Shannon's limit exactly. It quantifies the additional overhead beyond the theoretical minimum.

The first term corresponds to the information that remains secret from Eve after privacy amplification. Here, the single photon gain (G_1) represents the secure detections, since multi-photon pulses could be targeted by PNS attacks. The term $1 - H(\epsilon_1 Q_{BB84})$ can be interpreted as the amount of secret information that remains after accounting for errors and Eve's possible knowledge. The parameter ϵ_1 (heuristically decided) adjusts the observed QBER to account for the fact that eavesdropping on single-photon states may differ from the overall measured QBER. A factor of 1/2 represents the fact that in the standard BB84 protocol, exactly half of the raw bits are expected to be discarded during basis reconciliation.

5.2 | Decoy State QKD Protocol

The decoy state QKD, first suggested by Hwang et al. [60], was developed specifically for overcoming the Photon Number Splitting (PNS) attack in BB84. The PNS attack is particularly relevant in practical scenarios because weak coherent sources are used instead of perfect single-photon sources, which occasionally emit multiple photons per pulse. Eve exploits this vulnerability by intercepting one or more photons while allowing the others to reach Bob undisturbed, potentially extracting information while keeping the QBER relatively low. The existence of such vulnerabilities prompted the development of several modifications to the original BB84 protocol. In particular, one widely used variant is the *Weak + Vacuum Decoy State QKD* protocol, which enhances BB84 by adding extra signal states with different intensities (a weak decoy state and a vacuum state) in addition to the usual signal pulses, which helps detect eavesdropping more effectively. The steps involved are:

- i. Alice prepares a large number of optical pulses. Each pulse is randomly chosen to be one of three types: signal, decoy, or vacuum (no photon).
- ii. For each non-vacuum pulse, Alice also randomly selects a quantum state according to the BB84 encoding (i.e., one of two possible bases and one of two bit values).
- iii. Alice sends all the prepared pulses (signal, decoy, and vacuum) through the quantum channel to Bob.
- iv. Bob measures each incoming pulse by randomly choosing a measurement basis (one of the two BB84 bases).
- v. After transmission, Alice publicly announces which pulses were signal, decoy, or vacuum, and also which basis she used for encoding (but not the actual bit value).
- vi. Bob uses Alice's announcement to sort his detection results into three groups: signal results, decoy results, and vacuum results.
- vii. Alice and Bob use the vacuum results to estimate the dark count (background noise) of Bob's detectors.
- viii. They use the decoy results to estimate the detection rate and error rate of single-photon states, which are secure against photon-number-splitting attacks.

- ix. They then use only the signal results to generate the raw key.
- x. Alice and Bob apply error correction and privacy amplification on the raw key to obtain the final secure key.

5.2.1 | Decoy State QBER Analysis

This section describes the decoy state QKD protocol simulator based on the weak + vacuum decoy state protocol. This simulator relies heavily on mathematical analysis and some implied physical component modelling to describe the practical QKD system. The mathematical theoretical framework of the decoy state QKD protocol has been established in prior works like [17, 18, 61].

Combining the various physical components defined in the previous Section 3, we now define the inner workings of the simulator and model the decoy state QKD. In order to calculate the QBER, we decompose the transmitted coherent pulses into contributions from vacuum, single-photon, and multi-photon components. This allows a more manageable estimation of error rates and yields. Since a weak coherent pulse of average photon number (intensity) μ follows a Poisson distribution for the photon number, the contributions from vacuum ($n = 0$), single state ($n = 1$), and multi photon state ($n \geq 2$), respectively, are:

$$P_0 = e^{-\mu}, \quad P_1 = \mu e^{-\mu}, \quad P_{\geq 2} = 1 - P_0 - P_1 \quad (32)$$

Furthermore, the yield (Y_n) of these states is the probability that a signal containing n photons causes a detection at the receiver. Y_0 here would represent the background (dark count) probability, that is, the chance that Bob's detector clicks even when no photon arrives, due to detector noise or other errors. The single photon yield (Y_1) can be calculated as the sum of the chance of a real photon detection, which is represented by the total transmittance (η), and the chance of a background detection (Y_0). However, since both events could occur simultaneously (a real photon arrives while a dark count also triggers a click), we must subtract this overlap, given by the product (ηY_0). For pulses containing two or more photons, we approximate the multi-photon yield ($Y_{\geq 2}$) by the two-photon yield. Since the yield, in general, is monotone and increasing in n , this approximation serves as a conservative lower bound for all $n \geq 2$. The choice avoids the high computational cost of summing over all multi-photon terms, while still providing a security-consistent and slightly pessimistic estimate. For a detailed discussion of the mathematical derivations and the full decoy-state framework, see [17]. The yields are given as:

$$Y_1 = \eta + Y_0 - \eta Y_0$$

$$Y_{\geq 2} = 1 - (1 - \eta)^2 + Y_0 - Y_0(1 - (1 - \eta)^2) \quad (33)$$

The gain G_n is the joint probability of emitting an n photon pulse and detecting it, the general formula can be given as:

$$G_n = P_n Y_n \quad (34)$$

A vacuum pulse (0 photons) cannot trigger a detection unless a dark count (false positive) occurs in the detector. Since dark

counts are random, the resulting bit is equally likely to be 0 or 1. Therefore, the vacuum error probability is $\epsilon_0 = 1/2$. The single-photon error rate (ϵ_1) is a weighted sum of random background errors and intrinsic misalignment errors in the optical system, scaled by the probability that the photon successfully arrives at the detector. For multi-photon pulses, the error rate is approximated using an empirical scaling of the single-photon error rate. This reflects the fact that multi-photon states are more vulnerable to optical imperfections and potential adversarial strategies, such as PNS attacks, which increase the likelihood of erroneous detections. For flexibility and simplicity, we allow users to specify ϵ_1 as an input parameter; $\epsilon_{\geq 2}$ is then derived automatically through the scaling relation. This design enables straightforward exploration of different system conditions without adding unnecessary computational overhead. Once we estimate error rates for each component, we compute the total number of erroneous bits across all detections:

$$\epsilon_{\text{total}} = G_0\epsilon_0 + G_1\epsilon_1 + G_{\geq 2}\epsilon_{\geq 2} \quad (35)$$

Hence, the final QBER is the ratio of erroneous detections to total detections:

$$Q_{\text{decoy}} = \frac{\epsilon_{\text{total}}}{G_{\text{total}}} = \frac{G_0\epsilon_0 + G_1\epsilon_1 + G_{\geq 2}\epsilon_{\geq 2}}{G_0 + G_1 + G_{\geq 2}} \quad (36)$$

5.2.2 | Decoy State SKR Analysis

Next, we analyse the SKR of the decoy state QKD protocol, given by [17]:

$$SKR = q \left[G_1^L (1 - H_2(\epsilon_1^U)) - G_{\mu} f_{ec} H_2(Q_{\text{decoy}}^{\mu}) \right] \quad (37)$$

where $q = 1/2$ is the protocol efficiency, f_{ec} is the error correction efficiency, which is defined as the ratio between the actual number of bits revealed during error correction and the theoretical minimum number of bits needed (Shannon limit). In the simulator, $f_{ec} = 1.1$ is used as the default. G_{μ} is the overall gain for the signal state, and Q_{decoy}^{μ} is the average QBER, while $H_2(\cdot)$ is the binary Shannon entropy. Lastly, G_1^L is the lower bound gain of single-photon states, and ϵ_1^U is the upper bound error rate of single-photon states.

We begin by calculating the overall gain (G_{μ}), which is the total probability of detection of photons by Bob from Alice. It includes both real signal detections and background noise, given by:

$$G_{\mu} = Y_0 + (1 - e^{-\eta\mu}) \quad (38)$$

The overall gain for the decoy states can be calculated using Equation (38), where the weak decoy state has a mean photon number $\mu = \nu_1$, and the vacuum decoy state has a mean photon number $\mu = \nu_2 = 0$.

Using G_{ν_1} and G_{ν_2} , the lower bound of Y_0 or the background noise yield can be calculated as:

$$Y_0^L = \frac{\nu_1 G_{\nu_2} e^{\nu_2} - \nu_2 G_{\nu_1} e^{\nu_1}}{\nu_1 - \nu_2} \quad (39)$$

Similarly, we can calculate the lower bound of single-photon state yield (Y_1^L) and then the lower bounds of single-photon state gains (G_1^L), given by:

$$Y_1^L = \frac{\mu \left[G_{\nu_1} e^{\nu_1} - G_{\nu_2} e^{\nu_2} - \left(\frac{\nu_1^2 - \nu_2^2}{\mu^2} \right) (G_{\mu} e^{\mu} - Y_0^L) \right]}{\mu \nu_1 - \mu \nu_2 - \nu_1^2 + \nu_2^2} \quad (40)$$

$$G_1^L = Y_1^L \mu e^{-\mu} \quad (41)$$

Lastly, we also need to estimate the upper bound of the single-photon error rate. The upper bound of the error rate can be derived from the QBER difference of the two decoy states.

$$\epsilon_1^U = \frac{Q_{\text{decoy}}^{\nu_1} G_{\nu_1} e^{\nu_1} - Q_{\text{decoy}}^{\nu_2} G_{\nu_2} e^{\nu_2}}{(\nu_1 - \nu_2) Y_1^L} \quad (42)$$

A comprehensive discussion of the mathematical derivations and the full decoy-state framework can be found in [17].

5.3 | Coherent One-Way (COW) QKD Protocol

The COW QKD protocol, first discussed in [62], is based on the principles of time-bin photon entanglement. It is a widely implemented protocol due to its simplistic setup and ability to thwart PNS attacks. The protocol typically utilises a Weak Coherent Pulse (WCP) as the light source, to send a sequence of pulses to send the required bit.

The steps in the protocol are as follows:

- i. Alice prepares a sequence of coherent laser pulses, where each pulse has a fixed intensity.
- ii. Information is encoded into time-bin pairs. A logical '1' is represented by a filled pulse followed by an empty pulse ($|\alpha\rangle, |0\rangle$), while a logical '0' is represented by an empty pulse followed by a filled pulse ($|0\rangle, |\alpha\rangle$).
- iii. To ensure security, Alice also inserts decoy states at random positions. These decoy states consist of two consecutive filled pulses ($|\alpha\rangle, |\alpha\rangle$). They do not contribute to the final key but are used later to detect the presence of an eavesdropper by checking for decoherence effects.
- iv. The sequence of pulses (key bits and decoys) is transmitted through the quantum channel to Bob.
- v. Bob measures the incoming pulses using a detector system that is split into two parallel lines by a beam splitter, as shown in Figure 1:
 - a. Data line: This line functions as the main receiver of the raw key. Since Alice encodes each bit in two time slots, Bob only needs to check the arrival time of the pulse. If a pulse is detected in the first slot, the bit is '1'. If the pulse is in the second slot, the bit is '0'.

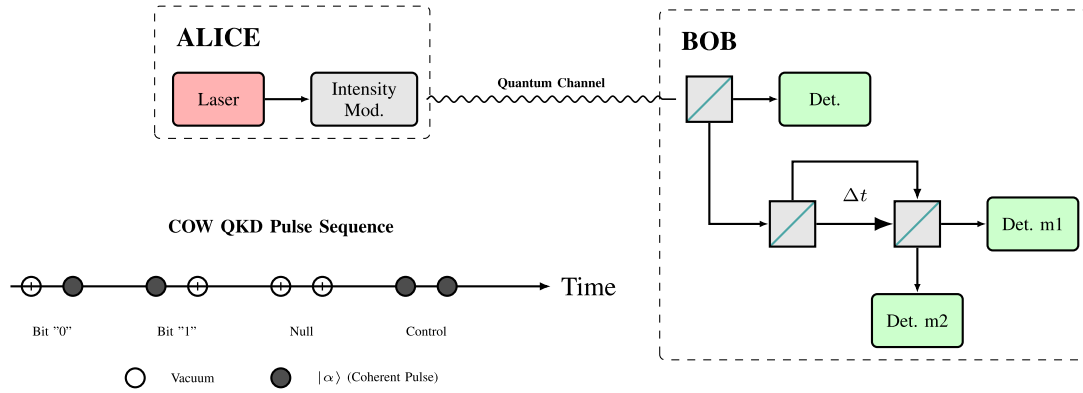


FIGURE 1 | This diagram illustrates the implementation of a Coherent One-Way (COW) QKD protocol. The protocol encodes information using pairs of time slots with either vacuum states or coherent pulses ($|\alpha\rangle$). The bit value '0' is encoded as ($|0\rangle, |\alpha\rangle$), bit '1' as ($|\alpha\rangle, |0\rangle$). Furthermore ($|0\rangle, |0\rangle$) represents a null sequence, and ($|\alpha\rangle, |\alpha\rangle$) is used as a control sequence to check the interferometric phase reference.

- b. Monitoring line: This line implements an interferometric setup that introduces a small fixed delay. This delay makes part of one pulse overlap with the next pulse, allowing interference between consecutive pulses. The monitoring line randomly selects a portion of the incoming pulses (including decoy states) and checks the coherence between successive filled pulses. If the quantum signal is untouched, high visibility interference is observed. If an eavesdropper attempts to measure or copy the pulses, this will disturb the states and reduce the observed visibility. Thus, the monitoring line does not contribute to the raw key but is essential for security checks.
- vi. After Bob's detection, Alice and Bob communicate over a classical authenticated channel to announce the positions of decoy states. They discard all events corresponding to decoy sequences.
- vii. For the remaining key bits, Alice and Bob perform key sifting, where they keep only those detection events where a single pulse was measured in the correct time slot.
- viii. Bob also calculates the visibility of the interference patterns in the monitoring line. This parameter quantifies how well quantum coherence was preserved during transmission. A high visibility indicates minimal disturbance, while a significant drop would suggest possible eavesdropping.
- ix. If both the error rate in the data line and the visibility in the monitoring line remain within secure thresholds, Alice and Bob proceed to the usual classical post-processing steps, namely error correction and privacy amplification, to generate the final secret key.

5.3.1 | COW QBER Analysis

Here we describe the inner workings of the COW QKD simulator. The implementation is inspired by prior works on COW protocol, such as [63–65]. Additionally, building upon the physical components, special attention should be made to Bob's detector system, where three identical detectors are used. One

for the data line to record the quantum bits and two for the monitoring line to perform security checks through interference measurements.

First, we consider the calculation of QBER; in order to do so, we need to define the unconditional probability of detecting a pulse at Bob's detector. This can be calculated using the mean photon number μ , and total transmittance η :

$$P_{\text{det}} = 1 - e^{-\mu\eta} \quad (43)$$

The QBER of the COW QKD simulator can be divided into two parts, the bit error component ($Q_{\text{cow}}^{\text{bit}}$) and the phase error component ($Q_{\text{cow}}^{\text{phase}}$). The $Q_{\text{cow}}^{\text{bit}}$ is explained by the dark count rates in the detector system, while the $Q_{\text{cow}}^{\text{phase}}$ accounts for imperfect interference visibility and is distance-dependent; it can be calculated as:

$$Q_{\text{cow}} = \frac{1}{2}(Q_{\text{cow}}^{\text{bit}} + Q_{\text{cow}}^{\text{phase}}) \quad (44)$$

The $Q_{\text{cow}}^{\text{bit}}$ is given by:

$$Q_{\text{cow}}^{\text{bit}} = \frac{P_{\text{dark}}}{P_{\text{det}} + P_{\text{dark}}} \quad (45)$$

where P_{dark} is the probability of a dark count event already defined in the detector systems, and P_{det} is the probability of a photon detection event, defined in Equation (43). The $Q_{\text{cow}}^{\text{phase}}$ is dependent on how well we can verify the coherence between consecutive pulses through interference measurements. The formula for calculating $Q_{\text{cow}}^{\text{phase}}$ is:

$$Q_{\text{cow}}^{\text{phase}} = \frac{1 - V_{\text{eff}} e^{-\mu/2}}{2} \quad (46)$$

where V_{eff} is the effective visibility, given by:

$$V_{\text{eff}} = \frac{V_0}{1 + \gamma d} \quad (47)$$

Which measures how effectively the coherent pulses can be differentiated from the incoherent ones. Here γ is the visibility

degradation coefficient, representing the rate at which interferometric visibility decreases with transmission distance due to phase noise, dispersion, and channel imperfections. A higher γ reduces the effective visibility V_{eff} , which increases $Q_{\text{COW}}^{\text{phase}}$ and therefore raises the overall QBER by weakening coherence verification between consecutive pulses. A value of $\gamma = 0.1\% \text{ km}^{-1}$ is commonly adopted in COW-QKD simulations as a conservative experimental estimate consistent with reported interferometric visibility decay [65]. In our simulator, the initial visibility is set to a typical value of ($V_0 = 8\%$) that deteriorates with distance (d), but this can be adjusted to model different system imperfections. The exponential term $e^{-\mu/2}$ accounts for the quantum coherence between consecutive pulses, and a factor of $1/2$ is included because the coherence is measured between two pulses.

5.3.2 | COW SKR Analysis

The SKR of COW protocol is given by:

$$\text{SKR} = q[R_{\text{raw}}(1 - f_{\text{ec}}H_2(Q_{\text{COW}}) - I_{\text{eve}})] \quad (48)$$

where R_{raw} is the raw key rate, which represents the rate at which potential key bits are generated before any security considerations, and $q = 1/2$ is the protocol efficiency. The raw key rate is calculated as:

$$R_{\text{raw}} = f_{\text{rep}} P_{\text{det}} \eta_{\text{data}} (1 - P_{\text{decoy}}) \quad (49)$$

Here, f_{rep} is the repetition rate, and η_{data} is the data line ratio, which is the percentage of the photon pulses used for the data line detectors, it is usually kept around 90%, while the remaining 10% goes to the monitoring line, where it gets tested for decoherence. The term P_{decoy} represents the probability of sending a decoy sequence. In our simulator, P_{decoy} is a user-defined parameter, allowing flexibility in testing different protocol settings. Next, we move to I_{eve} , which represents the amount of information that an Eve may gain about the raw key. It is calculated using two main factors, namely multi-photon leakage due to channel loss (r) and visibility-based leakage due to coherence loss (I_{vis}). r depends on the transmittance (η) and the mean photon number (μ) and can be formulated as $r = \mu(1 - \eta)$. Another source of information leakage arises from the reduced interference visibility in the monitoring line, given by the expression:

$$I_{\text{vis}} = \frac{1 - V_{\text{eff}}}{\xi}, \quad \xi = \frac{2e^{-\mu t}}{1 + e^{-\mu t}} \quad (50)$$

where ξ represents the probability that Eve detects a photon in one pulse but not in the next. This term links the security check (monitoring line interference) directly to Eve's possible information gain. Together I_{eve} can be formulated as:

$$I_{\text{eve}} = r + I_{\text{vis}}. \quad (51)$$

Lastly, the term $1 - f_{\text{ec}}H_2(Q_{\text{COW}})$ represents the fraction of the raw key that remains after correcting the errors introduced during transmission. Here f_{ec} is a heuristically decided

parameter, representing the efficiency of the error correction protocol used in practice. In order to learn more about the mathematical derivations and logic of COW protocol, readers are referred to [14, 64].

5.4 | Differential Phase Shift QKD Protocol

The Differential Phase Shift (DPS) QKD protocol uses the ability to encode information between the relative phase of two consecutive coherent pulses. This protocol was originally introduced in [66], where DPS QKD employs a train of sequential pulses with random phase modulation (typically 0 or π) applied between adjacent pulses, unlike COW QKD, where the information is encoded by the presence or absence of the pulse in specific time slots (time-bin encoding).

The DPS QKD protocol proceeds through the following steps (see Figure 2 for a schematic of the setup):

- i. Alice generates continuous coherent optical pulses ($|\alpha\rangle$) using a Weak Coherent Pulse (WCP) source. These pulses are attenuated to the single-photon level.
- ii. A random phase modulation (0 or π) is applied to the j^{th} pulse in the sequence, transforming the state to $|\alpha e^{i\phi_j}\rangle$.
- iii. The sequence of phase-modulated pulses is transmitted through a quantum channel to Bob. At Bob's side, the pulses are split into two paths using a delay Mach-Zehnder interferometric setup.
- iv. At the interferometer, the first beam splitter (BS1), splits the incoming pulses into two paths: the upper arm and the lower arm. The upper arm introduces a delay τ equal to the separation between two consecutive pulses.
- v. At the second beam splitter (BS2) in the interferometer, the delayed pulse from the upper arm interferes with the next consecutive pulse from the lower arm. This interference produces superposition states that are directed toward the two output ports.
- vi. The output ports of BS2 are connected to two single-photon detectors, D_0 and D_1 . Depending on the relative phase difference between consecutive pulses:
 - Detection at D_0 corresponds to bit value 0.
 - Detection at D_1 corresponds to bit value 1.
- vii. A detection event occurs only when a single photon is registered in a given time slot. The relevant time slots (TS1, TS2, etc.) correspond to overlapping pulse pairs such as TS1: pulse 1 (upper path) with pulse 2 (lower path), TS2: pulse 2 (upper path) with pulse 3 (lower path), and so on.
- viii. Bob publicly announces the time slots of successful detections. Alice, knowing her applied phase modulations, uses this information to reconcile the sifted key.
- ix. By comparing a subset of their bits, Alice and Bob estimate the error rate. If the error is within acceptable bounds, the sifted key is retained; otherwise, the presence of an eavesdropper (Eve) is inferred.

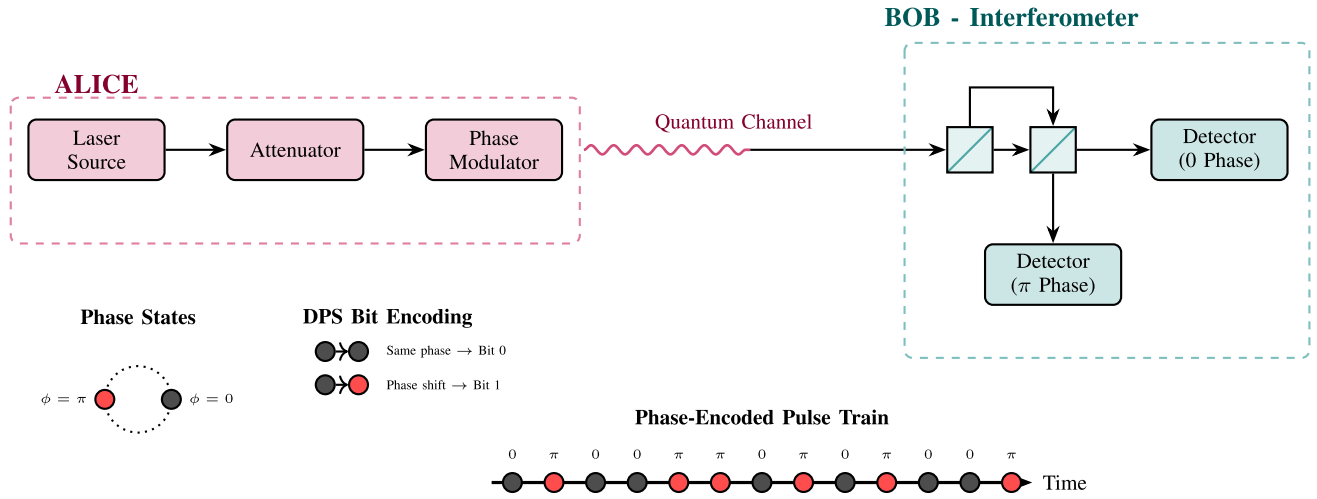


FIGURE 2 | In DPS QKD, Alice sends a sequence of light pulses with random phases (0 or π). The key is encoded in the phase difference between consecutive pulses: same phase = 0, different phase = 1. Bob measures these differences using a delay interferometer, where pulses interfere with the next one. Security comes from the fact that individual photons are hard to measure without disturbing the sequence.

- x. They estimate the error rate to detect potential eavesdropping. Since Eve cannot measure the phase difference between consecutive pulses without disturbing the quantum coherence, any interception will increase the error rate and reveal her presence.

5.4.1 | DPS QBER Analysis

In this section, we describe the DPS QKD simulator through mathematical formulae for the QBER and SKR analysis. Most of the technical mathematics involved can be studied further in papers like [23, 33, 67].

Building upon Section 3, two 50 : 50 beamsplitters are further modelled (BS1 and BS2) in Bob's interferometer. Firstly, we calculate the probability (yield) that Bob's detector clicks, conditional on receiving exactly n photons in a given time bin:

$$Y_n = \begin{cases} P_{\text{dark}}, & \text{for } n = 0 \\ 1 - (1 - \eta)^n, & \text{for } n \geq 1 \end{cases} \quad (52)$$

Using the aforementioned physical models, the QBER is formed as [68]:

$$Q_{\text{DPS}} = \frac{\frac{1}{2} P_{\text{dark}} + \epsilon_{\text{device}} P_{\text{signal}} + \frac{1}{4} P_{\text{multi}}}{P_{\text{signal}} + P_{\text{dark}} + P_{\text{multi}}} \quad (53)$$

where P_{dark} is the dark count probability per detection window.

The $1/2$ factor is introduced to account for the bit error, that is, due to the binary nature of the bits, there can only be two cases where the bits mismatch. The $1/4$ factor arises in multi-photon pulses, because Bob has only a 50% chance of a correct bit guess, and half of those events lead to errors. That gives an average 25% QBER contribution from multi-photon detections. The next term P_{signal} represents the probability, per detection window,

that a legitimate signal event is registered. A 'signal' event corresponding to the ideal case where a pulse containing exactly one photon is emitted is calculated as:

$$P_{\text{signal}} = \eta \mu e^{-\mu} \quad (54)$$

Finally, P_{multi} represents the probability of detecting events from pulses containing multiple photons, given as:

$$P_{\text{multi}} = \eta(1 - e^{-\mu} - \mu e^{-\mu}) \quad (55)$$

Next, ϵ_{device} the combined device error is calculated using:

$$\epsilon_{\text{device}} = \epsilon_{\text{vis}} + \epsilon_{\text{phase}} + \epsilon_{\text{jitter}} + \epsilon_{\text{pol}} + \epsilon_{\text{after}} \quad (56)$$

where ϵ_{vis} is the interferometer visibility, representing the imperfect interference of photons within the interferometer, given by:

$$\epsilon_{\text{vis}} = \frac{1 - V}{2} \quad (57)$$

where visibility (V) is a measure of the contrast of the interference fringes. In any real-world setup, factors such as imperfect mode matching of the interfering beams, unequal intensity in the interferometer arms, and component imperfections lead to $V \geq 1$.

The term ϵ_{phase} relates to the phase fluctuations in the interferometer, given by [69]:

$$\epsilon_{\text{phase}} = \frac{1}{2} \left(1 - e^{-\sigma_{\text{phase}}^2/2} \right) \quad (58)$$

where σ_{phase} denotes the standard deviation of phase fluctuations caused by environmental factors that alter the relative path lengths of the interferometer arms.

The timing jitter contribution, ϵ_{jitter} , originates from timing uncertainties in both the photon source and the detector. This contribution is modelled as the ratio of the detector's timing jitter t_{jitter} to the full duration of a bit window, which is twice the pulse width w_{pulse} . The pulse width is approximated from the system's repetition rate f_{rep} as $w_{\text{pulse}} \approx \frac{1}{2f_{\text{rep}}}$. The expression is calculated as:

$$\epsilon_{\text{jitter}} = \frac{t_{\text{jitter}}}{2w_{\text{pulse}}} \quad (59)$$

The polarisation error (ϵ_{po}) accounts for imperfections in maintaining the polarisation state of the photons as they travel through the optical system. Finally, (ϵ_{after}) represents the error contribution from detector afterpulsing.

5.4.2 | DPS SKR Analysis

The SKR of DPS QKD is given by [59]:

$$\text{SKR} = \frac{1}{2} \left[G_1 \left(1 - H_2 \left(Q_{\text{DPS}}^{\text{ph}} \right) \right) - G f_{\text{ec}} H_2 \left(Q_{\text{DPS}}^{\text{bit}} \right) \right] \quad (60)$$

where G_1 is the single-photon gain, G is the overall gain, and f_{ec} is the error correction efficiency as explained in previous sections. Additionally, $Q_{\text{DPS}}^{\text{ph}}$ and $Q_{\text{DPS}}^{\text{bit}}$ is the phase error and bit error respectively.

The $Q_{\text{DPS}}^{\text{bit}}$ is modelled as a summation of the probabilities of error from each distinct source. This approach assumes that the error mechanisms are largely uncorrelated. The composite formula is given by:

$$Q_{\text{DPS}}^{\text{bit}} = \epsilon_{\text{device}} + \epsilon_{\text{bg}} + \epsilon_{\text{elec}} \quad (61)$$

where ϵ_{device} is the same as given in Equation (53). The background noise error (ϵ_{bg}) represents the spurious detections caused by stray light or detector dark counts contribute to the QBER. These events are uncorrelated with the signal. The error probability is half the ratio of the background count rate (R_{bg}) to the total expected count rate (signal plus background):

$$\epsilon_{\text{bg}} = \frac{1}{2} \frac{R_{\text{bg}}}{R_{\text{sig}} + R_{\text{bg}}} \quad (62)$$

Here, the factor of 1/2 arises because a random background count has a 50% chance of occurring in the wrong bit value. Lastly, ϵ_{elec} represents the error originating from electronic noise in the control and detection circuitry.

The gain G_n can be calculated using the poisson probability $p(n)$ of emitting n photons with mean photon number μ , and the yield Y_n as:

$$G_n = p(n)Y_n \quad (63)$$

Hence, the total gain would be:

$$G = \sum_{n=0}^{\infty} G_n \quad (64)$$

Based on these yields and gains, we estimate the phase error rate $Q_{\text{DPS}}^{\text{ph}}$ through:

$$Q_{\text{DPS}}^{\text{ph}} = \frac{\frac{1}{2}G_0 + G_1\epsilon_{\text{bit}} + \sum_{n \geq 2} \frac{1}{4}G_n}{G} \quad (65)$$

where G_0 represents the contribution from the vacuum component, which leads to random errors assumed to occur with probability 1/2. Further, $G_1\epsilon_{\text{bit}}$ corresponds to the contribution from the single-photon component, with ϵ_{bit} denoting the bit error rate associated with these events. Lastly, $\sum_{n \geq 2} \frac{1}{4}G_n$ accounts for the multi-photon component under the worst-case assumption.

5.5 | E91 QKD Protocol

In 1991, Artur Ekert proposed a new QKD scheme grounded in the correlations of entangled particles and the violation of Bell's inequalities. The E91 QKD protocol [70] is structured thus:

- i. Entangled Particle Generation and Distribution: A source generates photon pairs in a maximally entangled singlet Bell state:

$$|\psi^-\rangle = \frac{1}{\sqrt{2}} (|H\rangle_A |V\rangle_B - |V\rangle_A |H\rangle_B), \quad (66)$$

where $|H\rangle$ and $|V\rangle$ represent horizontal and vertical polarisations. This state means that although each photon individually appears random, their measurement outcomes are perfectly anti-correlated when measured in the same basis. One photon is sent to Alice and the other to Bob through a quantum channel, such as an optical fibre or an FSO link.

- ii. Random Polarisation Measurement: Each user randomly chooses one of several polarisation measurement settings. Typically, Alice selects from $\{A_1, A_2, A_3\}$ and Bob from $\{B_1, B_2, B_3\}$. These settings correspond to specific polarisation angles designed to maximise the violation of the Clauser-Horne-Shimony-Holt (CHSH) inequality.
- iii. Data Collection: Alice and Bob record both the measurement basis and the outcome (horizontal or vertical polarisation) for each photon they receive.
- iv. Public Basis Comparison: Over an authenticated classical channel, Alice and Bob reveal a subset of their chosen measurement settings. This allows them to check correlations and estimate the level of security in the channel.
- v. Bell Test for Security: Using the subset of data, Alice and Bob calculate the CHSH Bell parameter S [50] to test for quantum correlations. A value of $S > 2$ demonstrates a violation of the inequality, confirming the presence of entanglement and ruling out classical eavesdropping.
- vi. Security Verification: If the Bell inequality is violated ($S > 2$), the channel is deemed secure, and the remaining undisclosed data can be used for key generation. If not, the protocol is aborted.

5.5.1 | E91 QBER Analysis

This section presents the quantitative modelling of QBER and SKR for the E91 entanglement-based QKD protocol, simulated under realistic channel and detector conditions.

5.5.1.1 | Accidental Coincidence Probability. The probability that two or more photon pairs are emitted in the same pulse thereby causing accidental coincidences. This is calculated by [50]:

$$\begin{aligned} P_{\text{accidental}} &\approx 1 - P(0) - P(1) \\ P_{\text{accidental}} &\approx 1 - e^{-\mu}(1 + \mu) \end{aligned} \quad (67)$$

Equation (67) captures the cumulative contribution of all $n \geq 2$ multi-pair emissions. Its value increases monotonically with μ , reinforcing the need to operate in the low-brightness regime to preserve entanglement fidelity and ensure secure key generation.

5.5.1.2 | Correlation Function and Bell Parameter Calculation. In the E91 protocol, the verification of entanglement and the security of the key rely on the violation of Bell inequalities specifically, the CHSH inequality. The core quantity used for this verification is the correlation function $E(\theta_A, \theta_B)$, which quantifies the expected correlation between measurement outcomes of entangled photon pairs when measured at angles θ_A and θ_B by Alice and Bob, respectively. The theoretical form of the correlation function for the singlet state $|\psi^-\rangle$ and a general entanglement phase ϕ is given by [50]:

$$E(\theta_A, \theta_B) = N[\cos(\phi) \sin(2\theta_A) \sin(2\theta_B) - \cos(2\theta_A) \cos(2\theta_B)] \quad (68)$$

where N is a normalisation factor accounting for imperfections such as noise and accidental coincidences, ϕ is the entanglement phase deviation from the ideal singlet state, θ_A and θ_B are the polarisation measurement settings of Alice and Bob, respectively. Here the correlation function measures how often Alice's and Bob's results agree or disagree for a given pair of measurement angles.

This formula arises from quantum mechanical predictions for joint detection probabilities under projective measurements in the polarisation basis. The sine and cosine components reflect the quantum interference terms expected from entangled states.

To test the CHSH inequality, the Bell parameter S is computed using four specific measurement settings:

$$S = |E(\theta_{A1}, \theta_{B1}) + E(\theta_{A1}, \theta_{B2}) - E(\theta_{A2}, \theta_{B1}) + E(\theta_{A2}, \theta_{B2})| \quad (69)$$

Here the Bell parameter acts as a proxy for channel quality, lower entanglement visibility translates into more mismatched key bits. These four angles are chosen based on the optimal violation of the CHSH inequality predicted by quantum mechanics. For the singlet state and ideal phase, the maximum quantum value of S is $2\sqrt{2}$, obtained when:

$$\theta_{A1} = 0, \quad \theta_{A2} = \frac{\pi}{4}, \quad \theta_{B1} = \frac{\pi}{8}, \quad \theta_{B2} = -\frac{\pi}{8} \quad (70)$$

These angles maximise the contrast between quantum predictions and classical local hidden variable theories, thus serving as a strong indicator of entanglement and nonlocality.

5.5.1.3 | Normalisation Factor and Detection Probability Modelling. To compute a reliable Bell parameter S_{CHSH} in realistic settings, we define a normalisation factor N that quantifies the ratio of trusted entangled detections to all two-photon coincidence events, including those affected by noise and multipair emissions. It is computed as [50]:

$$N = \frac{T^2 \eta^2 (1 - P_{\text{acc}})}{T^2 P_{\text{click}}^2 + 4 T(1 - T) P_{\text{noise}} P_{\text{click}} + 4 (1 - T)^2 P_{\text{noise}}^2} \quad (71)$$

where, T is the channel transmittance per arm, η is the total detection efficiency given by $\eta = \eta_d \eta_c$, where η_d denotes the intrinsic detector quantum efficiency and η_c represents the photon collection efficiency (typically $\eta_c = 0.6$). P_{noise} is the total noise probability per detector, encompassing both dark counts and external background light. The click probability per detector is given by $P_{\text{click}} = \eta + 2P_{\text{noise}}(1 - \eta)$, which includes both successful signal detections and spurious noise-induced clicks. The numerator of the normalisation factor N represents the probability that a transmitted photon from an entangled pair is detected at both ends, while excluding contributions from multi-pair emissions that can mimic genuine coincidences. The denominator comprises three distinct types of coincidence events: (i) both detectors successfully register photons, which may result from actual transmission, noise, or a combination thereof; (ii) only one detector receives a transmitted photon while the other registers a noise-induced click; and (iii) neither photon is transmitted to the detectors, but both register coincident clicks solely due to noise.

The QBER measures the fraction of incorrect bits in the sifted key shared between Alice and Bob. Instead of comparing key bits directly, QBER is estimated from the observed Bell parameter S , which quantifies the violation of the CHSH inequality. A lower S reflects degraded entanglement, leading to increased error rates. Under symmetric depolarising noise, the relationship is given by:

$$Q_{\text{E91}} = \frac{1}{2} \left(1 - \frac{S}{2\sqrt{2}} \right) \quad (72)$$

where $S \in [0, 2\sqrt{2}]$. This expression implies that perfect correlations ($S = 2\sqrt{2}$) yield $Q_{\text{E91}} = 0$, while values below the classical threshold $S \leq 2$ correspond to $Q_{\text{E91}} \geq 14.6\%$, indicating insecure transmission. Thus, QBER derived from Bell violations serves as a reliable metric for assessing the integrity of the quantum channel during key distribution.

5.5.2 | E91 SKR Analysis

The SKR quantifies the rate at which secure bits can be distilled from the raw key after accounting for both noise and potential eavesdropping. For the E91 protocol, the SKR is determined using a formulation proposed in [71], which connects the observed Bell violation with the amount of extractable secret

key under collective attacks. Provided a Bell parameter $S > 2$, the SKR is given by [50]:

$$SKR = \frac{1}{3} \mu f_{\text{rep}} T \left[1 - H_2(Q_{E91}) - H_2\left(\frac{1 + \sqrt{S^2/4 - 1}}{2}\right) \right] \quad (73)$$

where μ is the mean photon number per pulse, f_{rep} is the pulse repetition rate, T is the channel transmittance, and Q is the QBER.

The entropy term $H(Q_{E91})$ accounts for the information reconciliation cost, representing mutual information loss between Alice and Bob due to errors. The second entropy term captures the eavesdropper's potential knowledge and is used in privacy amplification to bound Eve's information. The prefactor $1/3$ reflects the basis mismatch probability inherent in the E91 protocol due to random measurement basis selection. When $S \leq 2$, the Bell inequality is not violated and no secret key can be generated ($R = 0$), indicating a fully insecure link.

5.6 | BBM92 QKD Protocol

Named after its creators, Bennett, Brassard and Mermin [53], the BBM92 is a popular variant of BB84 implemented with entangled photons. The BBM92 protocol can be described step by step as follows:

- i. Entangled-pair generation: Unlike BB84, where Alice prepares quantum states, BBM92 relies on entangled photon pairs generated by a source. Security arises from correlations between measurements rather than state preparation. A central source produces entangled photon pairs in the Bell-like state [47]. We use Equation (4), to represent this only here instead of μ , we utilise the parameter λ , which is one-half of the mean photon number μ . In BBM92, one photon from each generated pair is sent to Alice, and the other to Bob. (See Figure 3 for the schematics of the source and the two quantum channels.)
- ii. Distribution to the parties: The two photons travel through separate quantum channels (one toward Alice,

one toward Bob). Each party receives a stream of single photons from the source and directs them to their measurement apparatus (see Figure 3).

- iii. Random basis choice: For every incoming photon, Alice and Bob independently and randomly choose one of two measurement bases: the rectilinear (Z) basis or the diagonal (X) basis. Each party measures their photon in the chosen basis and records two items for each detection: (a) which basis was used (Z or X), and (b) the measurement outcome
- iv. Steps in post-processing (Steps 5–10) are the same as in BB84 (Section 5.1).

5.6.1 | BBM92 QBER Analysis

This section describes the various components of our simulator in order to calculate the QBER and SKR. For providing the mathematical and theoretical foundation of this simulator, we build upon the formulations established in [47].

The yield (Y_n) is defined as the conditional probability that a detection event occurs at both Alice's and Bob's side given that the PDC source emits an n -photon pair, given by [47]:

$$Y_n = [1 - (1 - Y_{0A})(1 - \eta_A)^n][1 - (1 - Y_{0B})(1 - \eta_B)^n] \quad (74)$$

where Y_{0A} , Y_{0B} are the background counts of Alice and Bob, they are assumed to be independent of single photon detection. The first bracket gives the probability Alice detects at least one photon (signal or background) from an n -pair emission; the second bracket gives the same for Bob. Multiplying gives the joint detection probability. The gain (G_n) of an n -photon pair, or the joint probability that the PDC source emits an n -photon pair and both Alice and Bob register a detection event, can be calculated using Equation (34). Thus, the overall gain G , which represents the total probability that a detection event occurs at both Alice's and Bob's side, regardless of the number of photon pairs emitted by the source, is given by [47]:

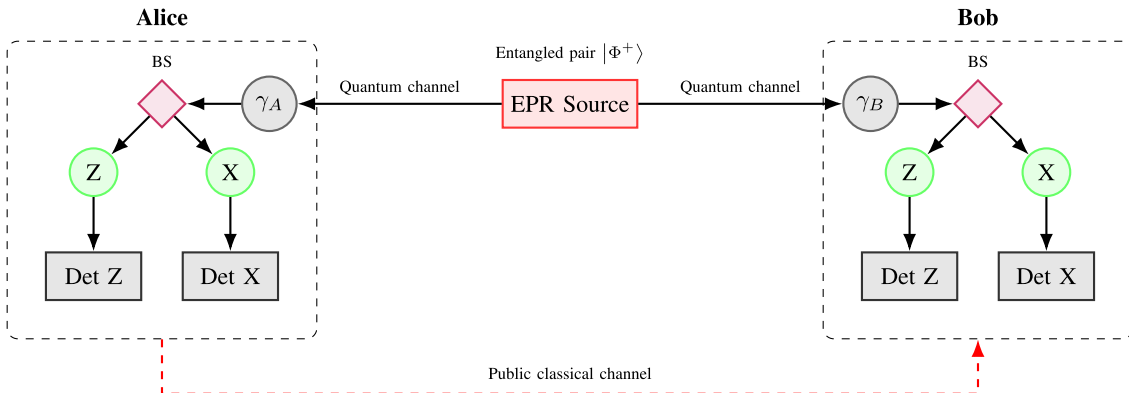


FIGURE 3 | This diagram illustrates the BBM92 protocol where an EPR source generates entangled photon pairs ($|\Phi^+\rangle$) distributed to Alice and Bob through quantum channels. Each party uses a beam splitter (BS) to randomly direct photons for measurement in either Z or X basis. After measurements, Alice and Bob share their basis choices through a classical public channel, keeping only results measured in matching bases to establish a secure key.

$$G = \sum_{n=0}^{\infty} G_n = 1 - \frac{(1 - Y_{0A})}{(1 + \eta_A \lambda)^2} - \frac{(1 - Y_{0B})}{(1 + \eta_B \lambda)^2} + \frac{(1 - Y_{0A})(1 - Y_{0B})}{(1 + \eta_A \lambda + \eta_B \lambda - \eta_A \eta_B \lambda)^2} \quad (75)$$

Finally, the QBER (Q_{BBM92}) is defined as [47]:

$$Q_{\text{BBM92}} = \epsilon_0 - \frac{2(\epsilon_0 - \epsilon_d)\eta_A\eta_B\lambda(1 + \lambda)}{G(1 + \eta_A\lambda)(1 + \eta_B\lambda)(1 + \eta_A\lambda + \eta_B\lambda - \eta_A\eta_B\lambda)} \quad (76)$$

where ϵ_0 and ϵ_d represent background counts and intrinsic detector errors in the QKD system. Specifically, ϵ_0 accounts for background counts, which are random detection events independent of the signal, while ϵ_d represents intrinsic detector errors arising from imperfect alignment, stability, or calibration of Alice's and Bob's measurement devices. In our implementation, ϵ_d is assumed to increase linearly with distance because longer optical channels amplify the effects of misalignment, dispersion, and other imperfections. Also, multi-photon-pair emissions and double-click events contribute additional errors, with double clicks assigned random bits, resulting in an error probability of $\epsilon_0 = 1/2$.

5.6.2 | BBM92 SKR Analysis

The SKR for the BBM92 QKD can be defined as [72]:

$$SKR \geq q \left\{ G \left[1 - f_{ec} H_2(Q_{\text{BBM92}}^{\text{bit}}) - H_2(Q_{\text{BBM92}}^{\text{ph}}) \right] \right\} \quad (77)$$

where $q = 1/2$ is the basis reconciliation factor, f_{ec} is the error correction efficiency, which accounts for the overhead in practical error correction compared to the ideal Shannon limit (by default f_{ec} is set to 1.1). $Q_{\text{BBM92}}^{\text{bit}}$ and $Q_{\text{BBM92}}^{\text{ph}}$ are the bit and phase error rate, respectively. Due to the symmetry of X and Z bases measurements $Q_{\text{BBM92}}^{\text{bit}} = Q_{\text{BBM92}}^{\text{ph}} = Q_{\text{BBM92}}$.

6 | Numerical Simulation

Numerical simulations are important in order to measure the SKR and QBER of QKD protocols under realistic optical conditions. They allow researchers and learners to explore, test, and understand protocol performance before working with physical instruments or costly experimental setups. The simulation results are presented through plots that illustrate the relationship between SKR and QBER in both fibre and free-space channels. Table 4 summarises the parameter values used in the simulation for a consistent comparison across different QKD protocols. In the plots where one of the parameters (such as distance or μ) is varied, the remaining parameters are held constant at the values given in Table 4.

Figure 4a,b demonstrate the relation between metrics like QBER and SKR to parameters like distance and mean photon numbers in a fibre-based channel. The parabola-shaped curve of

QBER in Figure 4a indicates an optimal operating region between $\mu = 0.1$ and $\mu = 0.2$. At very low μ values, dark counts dominate the detection events. As μ increases beyond the optimal region, the increasing multi-photon pulse contributions dominate and thus QBER increases.

On the other hand, SKR shows an inverse relation, with a peak at around $\mu = 0.1$ when compared to QBER. Figure 4b shows the QBER increasing with distance and SKR decreasing exponentially as the transmission distance increases. This behaviour can be attributed to channel losses and dark count contributions. Figure 4c,d show a similar relation when using an FSO-based channel. There are higher losses in the FSO-based channel compared to the fibre-based channel, due to beam divergence, atmospheric losses, etc. As a result, the FSO-based channel exhibits higher QBER and reduced SKR over similar transmission distances. These observed limits and performance trends align closely with established experimental benchmarks for the BB84 protocol [73].

Figure 4e-h illustrate the relation between SKR, QBER and transmission distance, mean photon number for Decoy State Protocol, analogous to the graphs for the BB84 protocol. A lot of similar trends are also observed in both of these figures, with a general increase in SKR for the same transmission distance and mean photon number due to improvements in security and efficiency due to the use of decoy states. These values and trends are consistent with foundational theoretical models for decoy-state implementations [17].

Figure 5a-d show the plots of the SKR and QBER versus distance for the COW QKD protocol. Here, we can see a significant increase in overall SKR. Unlike the aforementioned protocols, COW leverages time-bin encoding and coherence between successive pulses, which enhances photon utilisation and allows higher key generation rates and security over longer distances. These SKR and QBER figures accurately reflect the theoretical expectations for the COW protocol [65].

Figure 5e-h, similar to previous graphs, exhibits a consistent inverse relationship between SKR, QBER and transmission distance for the DPS protocol. It shows a performance similar to the COW protocol with comparable SKR and QBER for the same distance and mean photon number. While its overall trends are close to those of the COW protocol, the DPS protocol achieves this through phase-encoded key generation, giving us a reliable alternative. These trends are corroborated by well-known experimental evaluations of the DPS protocol [74].

Figure 6b,c illustrate the relation between QBER, SKR and mean photon number, transmission distance for E91, an entanglement-based QKD protocol. We observe a higher SKR rate for corresponding values. However, the SKR decays faster with respect to transmission distance. This faster decay is due to the higher sensitivity of entangled photons to losses and decoherence over longer distances [50].

Figure 6e-h demonstrate the relationship between QBER, SKR, mean photon number, and transmission distance for BBM92 [57], another entanglement-based QKD protocol. The results confirm the overall trend observed in the aforementioned QKD

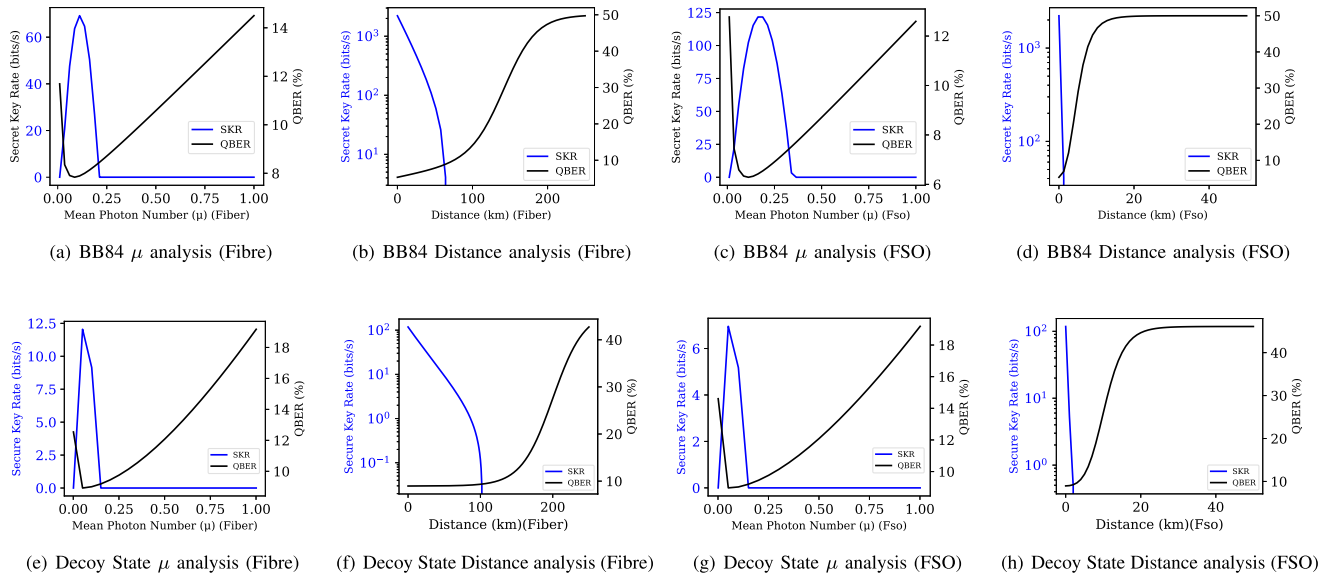


FIGURE 4 | Simulation graphs for BB84 (a)–(d) and Decoy State (e)–(h) protocols. These plots illustrate how the QBER and the SKR vary with changes in the mean photon number (μ) and transmission distance in each communication mode.

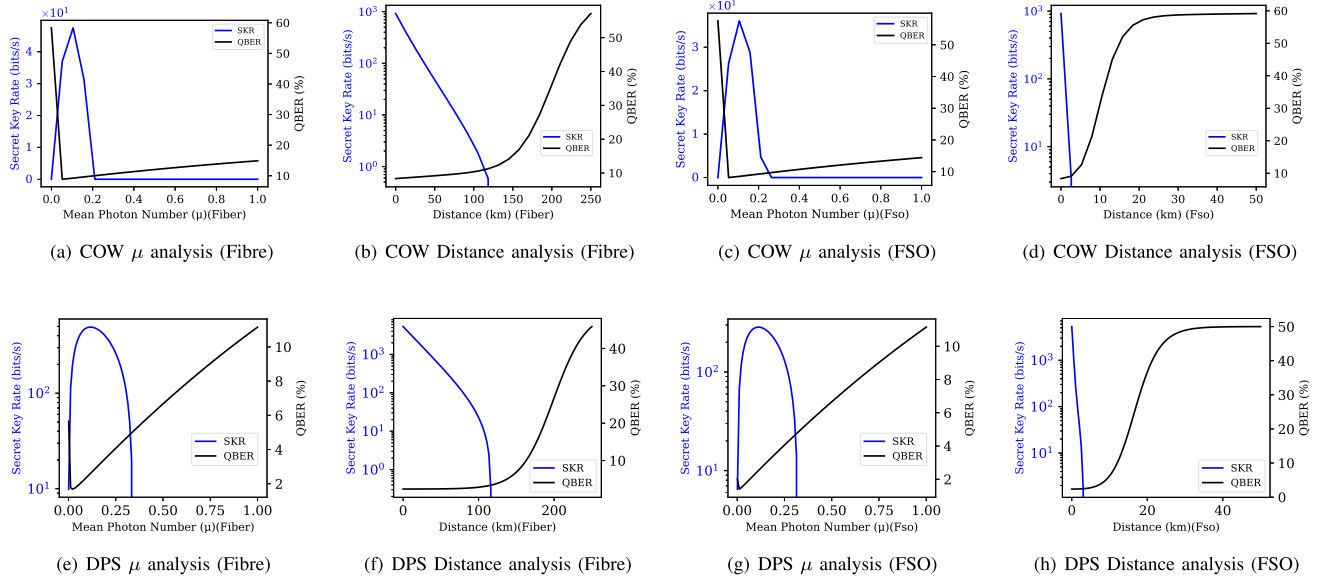


FIGURE 5 | Simulation graphs for COW (a)–(d) and DPS (e)–(h) protocols. These illustrate how the QBER and the SKR vary with changes in the mean photon number (μ) and transmission distance in each communication mode.

protocols. The security is seen to be slightly weaker than that of E91.

Figure 7 presents the SKR trends for all the aforementioned protocols versus the transmission distance in both Fibre and FSO mode. From the plots, we can observe that in Fibre mode, protocols like E91, COW, and DPS have a much higher SKR compared to the others. However, in terms of transmission range, COW and DPS outperform E91 and other protocols. There is the obvious difference between the entanglement-based protocols and prepare-and-measure-based protocols, that is entanglement-based protocols have a relatively lower transmission range due to the sensitivity of the entangled photons. COW and DPS protocols also outperform BB84, Decoy State

QKD protocol due to their phase-encoded design, which provides higher tolerance to photon loss and improved stability over longer distances. A similar hierarchy among the protocols is observed in FSO mode as well, although the overall transmission range is reduced due to higher channel losses and atmospheric turbulence inherent in free-space optical links.

7 | Graphical Interface

The graphical user interface (GUI) is implemented as a web-based application utilising the Python Django framework. The primary method of accessing the simulator is through a

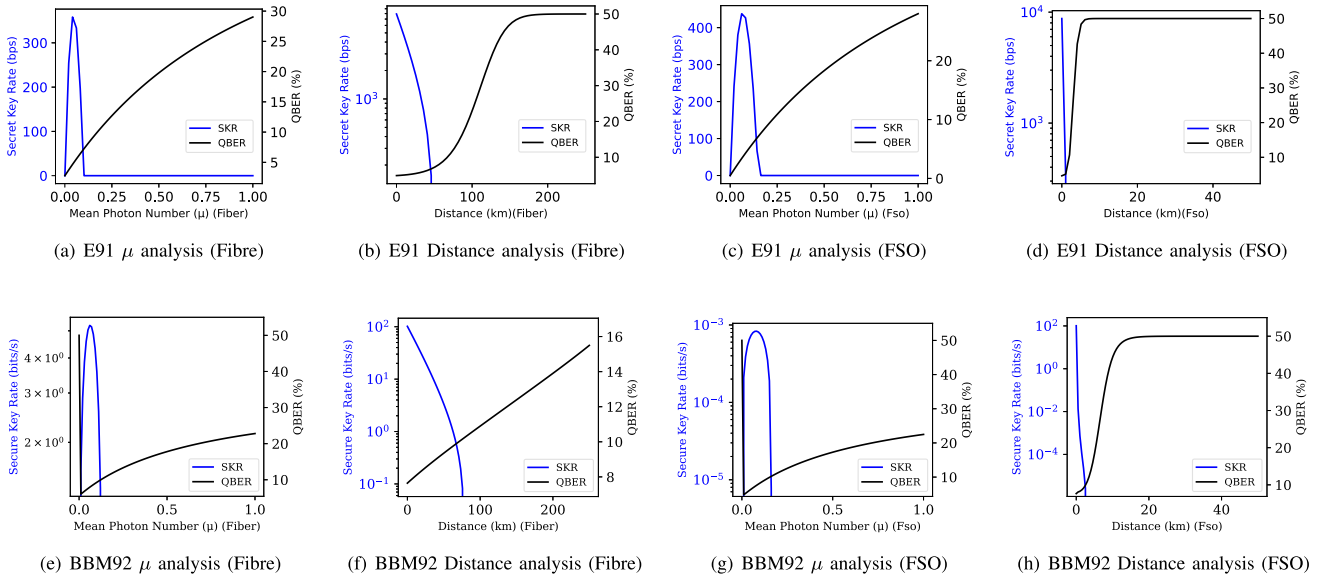


FIGURE 6 | Simulation results for the entanglement-based E91 (a)–(d) and BBM92 (e)–(h) protocols. These plots illustrate how the QBER and the SKR vary with changes in the mean photon number (μ) and transmission distance in each communication mode.

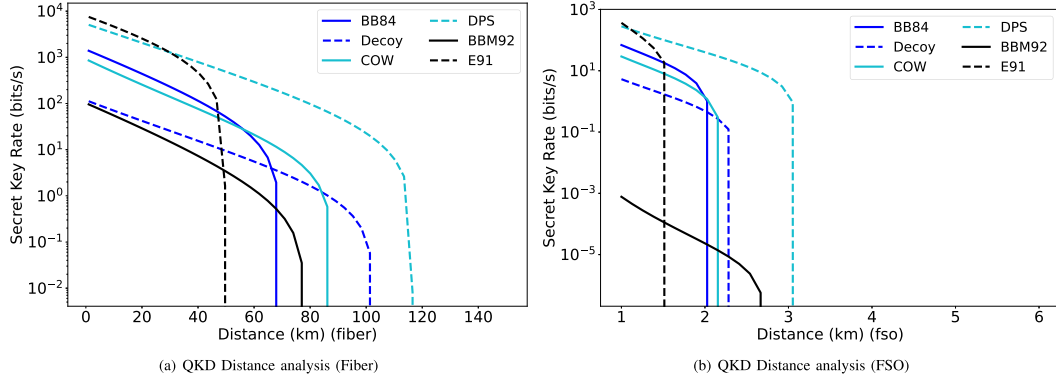


FIGURE 7 | The SKR of six different QKD protocols (BB84, Decoy, COW, DPS, BBM92, E91) is compared. The plots show the decline in the secret key rate (bits/s) as the transmission distance increases in both optical fibre (a) and free-space optical (FSO) links (b).

standard web browser, which provides a straightforward, platform-independent environment for analysis (see Figure 8). As an alternative to the web-based deployment, the software can also be executed locally. To facilitate this, we provide standalone executables for both Linux and Windows operating systems. These local installations allow users to run the simulations in an offline environment without the need to manually configure the underlying software dependencies. The code and software that support the findings of this study are available from the corresponding author upon reasonable request.

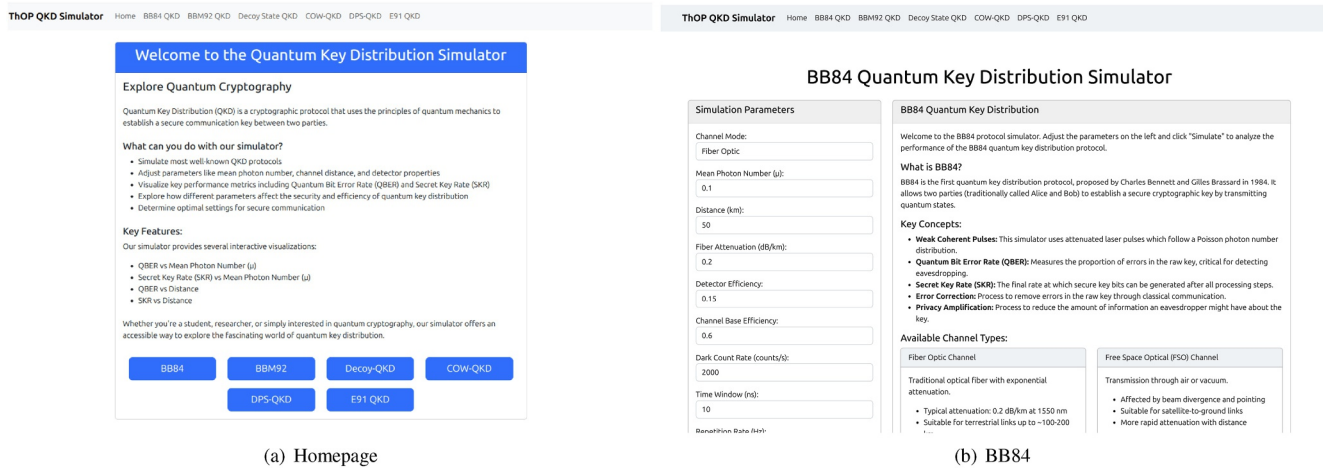
8 | Conclusion

This tutorial presents a detailed review of various QKD protocols, including BB84, Decoy State, COW, DPS, BBM92, and E91. To evaluate their performance, we develop a hybrid analytical–simulation framework that implements protocol-specific QBER and SKR models. This framework incorporates physical effects such as channel loss, detector inefficiency, dark counts, and eavesdropper interventions, enabling calculation of the QBER and the resulting SKR as functions of system

parameters. Our results show how each protocol’s design (basis choices, decoy intensities, etc.) leads to characteristic performance trade-offs.

This approach confirms and reproduces theoretical deductions, for instance, QBER remains low under short, clean channels but rises rapidly with loss and noise, and highlights differences between protocols like the inverse relation between QBER and SKR, and their correlation between mean photon number and channel distance.

Beyond this, we have also focused on visualisation and accessibility by delivering an interactive web-based GUI. The interface provides beginners and learners a platform to interact with and allows an intuitive understanding of QKD and enables them to observe the immediate impact of parameter changes on QBER and SKR. Despite these advantages, several limitations should be noted (see Table 3). The framework currently focuses on discrete-variable (DV) QKD protocols, and continuous-variable (CV) schemes are not yet implemented, which limits comparisons across the full range of QKD approaches. Furthermore, eavesdropping effects are modelled using simplified parameters



(a) Homepage

(b) BB84

FIGURE 8 | The images show the graphical user interface of the QKD simulator for different protocols. They display input parameter panels, key performance results, and plots of Quantum Bit Error Rate (QBER) and Secret Key Rate (SKR).

TABLE 3 | Summary and comparison of the investigated QKD protocols.

Protocol	Category	Encoding scheme	Key mechanism & security	Performance characteristics
BB84	Prepare-and-Measure	Polarisation (Z and X bases)	Information is encoded in mutually non-orthogonal bases. Vulnerable to photon number splitting (PNS) attacks when using weak coherent pulses (WCP).	Baseline secret key rate (SKR); performance and transmission range are limited by multi-photon errors and losses.
Decoy state	Prepare-and-Measure	Polarisation + intensity variation	Introduces weak decoy and vacuum states to detect eavesdropping. Specifically designed to thwart PNS attacks.	Higher SKR and improved transmission range compared to standard BB84 due to better multi-photon error handling.
COW	Prepare-and-Measure	Time-bin (coherent pulses)	Uses pulse position (empty/filled) for data and an interferometric monitoring line to check for decoherence. Highly resistant to PNS attacks.	High SKR with a significantly longer transmission range. Robust against photon loss.
DPS	Prepare-and-Measure	Phase difference (0 or π)	Encodes bits in the relative phase between consecutive pulses. Bob uses a delay Mach-Zehnder interferometer to detect phase shifts.	Performance is comparable to COW; offers high SKR and extended transmission distances.
E91	Entanglement-based	Bell states (maximally entangled)	Security is fundamentally verified by testing for quantum correlations via the violation of the CHSH (Bell's) inequality ($S > 2$).	Extremely high SKR at short distances, but rate decays rapidly over distance due to the fragility of entangled states.
BBM92	Entanglement-based	Entangled photon pairs ($ \Phi^+\rangle$)	Operates similarly to BB84 but relies on a central EPR source emitting entangled photons to Alice and Bob.	Follows similar range limitations as E91. Security threshold is slightly weaker compared to the E91 protocol.

embedded within the SKR formulation rather than explicit attack simulations. While non-ideal behaviour of sources, detectors, and channels is included through probabilistic efficiency and noise parameters, these models remain abstracted and do not capture all hardware-specific imperfections. These simplifications enable computational efficiency and protocol-level analysis but represent important limitations that future extensions aim to address. Another key limitation of the current simulator is that all results are evaluated in the asymptotic limit and therefore do not include

finite-key statistical effects [64, 75, 76]. In practical QKD systems, finite data size introduces additional security penalties that reduce achievable secret key rates. Extending the framework to include finite-key analysis, confidence bounds, and realistic parameter estimation procedures represents an important direction for future work. Furthermore, the simulator reproduces well-established theoretical trends reported in the QKD literature; but is not calibrated against a specific experimental platform. Experimental implementations often differ in hardware

TABLE 4 | Simulation parameters for QKD protocols.

Description	Value
Source wavelength (λ)	1550nm
Mean photon number per pulse (μ)	10^{-1}
Transmission distance (fibre)	50 km
Transmission distance (FSO)	1 km
Detector efficiency	10^{-1}
Dark counts per second	10^3
Dark count probability per pulse	10^{-6}
Time window duration	10^{-8} s
Fibre attenuation coefficient	0.2 dB/km
Atmospheric attenuation coefficient	0.2 dB/km
Transmitter aperture diameter	0.1 m
Receiver aperture diameter	0.3 m
Beam divergence angle	10^{-3} rad
Channel base efficiency	1
Pulse repetition rate	10^6 Hz

components, detector characteristics, loss models, and post-processing methods, which makes direct one-to-one comparison difficult [32, 35]. Therefore, the present work focuses on theoretical consistency and controlled relative comparison across protocols rather than exact experimental replication.

Author Contributions

Ushnik Nath: conceptualization, data curation, investigation, methodology, software, validation, visualization, writing – original draft, writing – review and editing. **Ramniwas Meena:** data curation, formal analysis, methodology, visualization, writing – original draft, writing – review and editing. **Shahnoor Ali:** conceptualization, formal analysis, methodology, validation, visualization, writing – review and editing. **Neel Kanth Kundu:** conceptualization, formal analysis, funding acquisition, investigation, project administration, resources, supervision, validation, writing – original draft, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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